

# PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/Second Semester/Final  
Time: 03:00 hrs. Full Marks: 60/Pass Marks: 24

**MCA192: Artificial Intelligence** (New Course)

Candidates are required to give their answers in their own words as far as practicable. Figure in the margin indicate full marks.

## Group A

2×12=24

**Answer TWO questions.**

1. What is searching and what are its different techniques? Compare the algorithms, advantages and disadvantages of Depth-First Search and Breadth-First Search with suitable examples of a tree traversal. 2+10
2. Define logic and tautology. State and explain Inference Rules with suitable examples. 4+8
3. What is Artificial Neural Network? Explain the structure of an ANN. Briefly describe Hopfield network and its applications. 2+6+4

## Group B

6×6=36

**Answer SIX questions.**

4. Define PEAS and discuss about the types or properties of environments.
5. What do you mean by adversarial search? What is Min-Max problem and how does Alpha-Beta Pruning simplifies the problem?
6. Convert the following sentences into FOPL:  
(a) Mohan likes to swim on sunny days.  
(b) There is no one who doesn't want to earn.  
(c) Some teachers are intelligent.
7. Solve the following cryptarithmic problem using Constraint Satisfaction, stating necessary assumptions that you make.

$$\text{RAJA} + \text{RANI} = \text{MAHAL}$$

8. What are the major categories of knowledge and what are knowledge elicitation techniques in Expert system?
9. What is the importance of learning in AI? Explain the working of Genetic algorithm.
10. What is Natural Language Processing? Construct a parse-tree for the following sentence:  
The disobedient student misbehaved with new teacher.
11. Write short notes on Any TWO: 2×3=6  
(a) Reinforcement learning (b) History of AI (c) Deep learning

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# PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/Second Semester/ Final  
Time: 03:00 hrs. Full Marks: 60/Pass Marks: 24  
**MCA193: Machine Learning (Specialization-II) (New Course)**

Candidates are required to give their answers in their own words as far as practicable. Figure in the margin indicate full marks.

## Group A

Answer TWO questions.

2×12=24

1. What do you mean by learning models? Explain logic model and geometric model with examples. Briefly discuss about the steps of designing a learning system. 2+6+4
2. Define entropy and information gain. Construct decision tree for the following: 2+10

| Day | Outlook  | Temperature | Humidity | Wind   | Decision |
|-----|----------|-------------|----------|--------|----------|
| 1   | Sunny    | Hot         | High     | Weak   | Yes      |
| 2   | Sunny    | Hot         | High     | Strong | No       |
| 2   | Overcast | Hot         | High     | Weak   | Yes      |
| 4   | Rain     | Mild        | High     | Weak   | No       |
| 5   | Rain     | Cool        | Normal   | Weak   | Yes      |

3. Analyze how overfitting occurs in decision tree and describe methods such as pruning and setting a maximum depth to mitigate this issue.

## Group B

Answer SIX questions.

6×6=36

4. What is the difference between binary and multiclass classification?
5. Explain K-means clustering algorithm.
6. Explain how Support Vector Machine (SVM) works for classification task. What are key parameters that need to be tuned for SVM models?
7. Considering a long list of machine learning algorithms, given a data set, how do you decide which one to use? Justify your answer.
8. Explain K-fold Cross-Validation in machine learning.
9. Explain the importance of text mining in machine learning.
10. What is an activation function in an artificial neuron? Give some examples.





# PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/Second Semester/ Final  
Time: 03:00 hrs. Full Marks: 80/Pass Marks: 32

**MCA185: Digital Commerce (Specialization-I) (New Course)**

Candidates are required to give their answers in their own words as far as practicable. Figure in the margin indicate full marks.

## Group A

Answer TWO questions.

2×12=24

1. Define business model. Explain in detail about different business models of E-commerce. 2+10
- 2(a) Compare E-commerce, Digital commerce and Mobile commerce. 6
- (b) Write down the challenges of Digital commerce. 3
- (c) Briefly explain the applications of M-commerce. 3
- 3(a) What are the major differences between traditional marketing and digital marketing? 4
- (b) What is display marketing? Explain social media marketing and mix marketing with their impact in modern business. 2+6

## Group B

Answer SEVEN questions.

7×8=56

4. What do you mean mercantile model? Explain distinct phases of a customer mercantile model.
5. What are the wireless devices and technologies that turn E-business into M-commerce? Explain their features.
6. How does a WCMS works? Explain the role of dynamic web content sites in digital commerce.
7. What is MOODLE? Explain its features and advantages.
8. How can Artificial Intelligence help business optimize their digital commerce strategies?
9. What are the most commonly used digital tokens in online transactions today? Explain in brief.
10. Compare the features of 3G, 4G and 5G.
11. Write short notes on Any TWO:  
(a) SEO (b) Firewall (c) GSM 2×4=8





# PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/Second Semester/Final  
Time: 03:00 hrs. Full Marks: 80/Pass Marks: 32

**MCA186: Digital Governance (Specialization-II) (New Course)**

Candidates are required to give their answers in their own words as far as practicable. Figure in the margin indicate full marks.

## Group A

2×12=24

**Answer TWO questions.**

1. List and describe two core components of e-government systems. Discuss how e-government functions as an information system. Analyze its architecture, including data management, user interfaces, and communication channels. Provide examples to illustrate how these components interact to improve government operations.
2. Discuss the essential infrastructure components necessary for the successful implementation of e-governance systems. Include considerations for hardware, software, networking, and human resources in your analysis.
3. Discuss the key security challenges faced by e-governance systems. Include issues such as data breaches, cyber-attacks, and insider threats in your analysis. How do these challenges impact the effectiveness and trustworthiness of e-governance services?

## Group B

**Answer SEVEN questions.**

7×8=56

4. Define the term Public-Private Partnership (PPP) and provide an example of a PPP model used in e-governance.
5. Define e-readiness. Briefly explain the steps to e-Government readiness.
6. What are some emerging management issues in the implementation of e-government systems? Provide two examples.
7. What are the key components of a project management plan for an e-government project?
8. What is the role of risk identification in the risk assessment process for e-governance?
9. What do you mean by digital policy and what are its attributes?
10. What are digital standards, and why are they important for digital governance and technology implementation?
11. Write short notes on Any TWO:  
(a) WWW Governance (b) Data centers in Nepal (c) Digital presence



# PURBANCHAL UNIVERSITY

**2024**

Master of Computer Application (M.C.A.)/Second Semester/*Final*

Time: 03:00 hrs.

Full Marks: 60/Pass Marks: 24

**MCA153: Cloud Computing** (New Course)

*Candidates are required to give their answers in their own words as far as practicable. Figure in the margin indicate full marks.*

## Group A

**2×12=24**

**Answer TWO questions.**

1. Assess the trade-offs between security and convenience in cloud computing. How can organizations strike the right balance between these two aspects? Explain.
2. Analyze the benefits and challenges of adopting a hybrid or multi-cloud strategy. How can organizations effectively manage workloads across multiple cloud providers or environments? Evaluate the role of containerization and orchestration technologies like Kubernetes in enabling hybrid and multi-cloud deployments.
3. Explore the concept of server virtualization in depth. How does it work, and what are the benefits of consolidating multiple virtual machines (VMs) on a single physical server? What are the challenges and best practices for managing server virtualization infrastructure? Explain.

## Group B

**6×6=36**

**Answer SIX questions.**

4. List few cloud vendors. Explain few security issues and other challenges faced by their vendors in implementing cloud services.
5. Compare the cost benefits of a public cloud, private cloud, and hybrid cloud deployment model for a medium-sized enterprise. What factors should drive the decision-making process?
6. What is scalability in cloud computing? how can businesses design their cloud infrastructure to ensure both horizontal and vertical scalability?
7. Consider the lock-in effect when choosing a cloud provider. How can organizations make strategic decisions to minimize vendor lock-in and maintain flexibility?

**Contd. ...**

(2)

8. Analyze the cost implications of different disaster recovery strategies in the cloud, such as backup and replication. What trade-offs exist between cost and recovery time objectives?
9. Analyze the legal and ethical considerations related to data privacy, and sovereignty in the context of cloud computing. How can organizations navigate these complex issues? Give your view.
10. Investigate the relationship between virtualization and cloud computing. How does virtualization underpin the infrastructure as a service (IaaS) model in the cloud, and how do cloud providers leverage virtualization to offer scalable resources to customers?
11. Write short notes on Any TWO:  
(a) Big Data and Big Data Analytics  
(b) Service Level Agreements (SLAs)  
(c) Grid Computing standards and architecture

2×3=6





**PURBANCHAL UNIVERSITY****2024**Master of Computer Application (M.C.A.)/Second Semester/ *Final*

Time: 03:00 hrs.

Full Marks: 60/Pass Marks: 24

**MCA154: Information Security and Ethics** (New Course)*Candidates are required to give their answers in their own words as far as practicable. Figure in the margin indicate full marks.***Group A****Answer TWO questions.****2×12=24**

1. Illustrate the Diffie-Heillman Key Exchange protocol with a suitable example.
2. Briefly write about the threats in network. Discuss about the security protocols of different layers in TCP/IP model. 4+8
3. What are the controls against program threats? Throw lights on the role of Intrusion Detection and Prevention System in controlling threats and malicious programs. 3+9

**Group B****Answer SIX questions.****6×6=36**

4. What are the requirements of a good encryption algorithm? Explain.
5. Discuss the security of One Time Pad with suitable illustrations.
6. Explain about several mechanisms of file protection.
7. What should be the organizational security policies in protecting physical resources of an organization? Write about issues that arise in the administration level.
8. Exemplify the Radix-64 conversion in context of Pretty Good Privacy.
9. Describe in short the taxonomy (various categories) of computer crimes.
10. When does a Copyright of an intellectual property begin and expire? Discuss.



# PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/Second Semester/Final

Time: 03:00 hrs.

Full Marks: 80/Pass Marks: 32

**MCA151/MCA121: Research Methodology (New/Back)**

*Candidates are required to give their answers in their own words as far as practicable. Figure in the margin indicate full marks.*

## Group A

**Answer TWO questions.**

**2×16=32**

1. What is the importance of research? List and describe the steps of scientific research process.
2. What are the features of good research question? Explain the types of hypotheses with examples.
3. What are the methods of sampling? What are the sources of primary and secondary data? Mention the types of coding.

## Group B

**Answer EIGHT questions.**

**8×6=48**

4. Explain features of qualitative, quantitative and mixed research method.
5. What is literature review? What are the steps of writing literature review? Explain.
6. What are ethical issues in research? Provide examples of plagiarism.
7. What do you understand by research design? Distinguish between qualitative and quantitative research design.
8. What is the importance of data reduction? Discuss any three selected techniques of data analysis.
9. Explain the basic steps of writing research proposal.
10. What are the issues or concerns in conducting observational research? Briefly mention the case study method of research.
11. What is the difference between the dependent variable and the independent variable? What problem do experiments often have?





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Time: 03:00 hrs.

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**MCA152/MCA122: Visual Programming Language and .NET (New/Back)**

Candidates are required to give their answers in their own words as far as practicable. Figure in the margin indicate full marks.

## Group A

Answer TWO questions.

2×12=24

1. What is a dialog box? Explain its types. What is the problem of using a modeless dialog box? Suggest ways to solve those problems. 2+4+3+3
2. What is a gizmo? What are the various phases of direct manipulation? Discuss the various direct-manipulations techniques. 2+4+6
3. What is master page in ASP.Net? Design the following webpage using the concept of master page. Link all the menu items to the {Placeholder} as shown in the figure below. Your page should also have placeholder in the title part of the webpage. 7+3+2

| Purbanchal University                                                                     |          |                                                                                    |        |        |       |
|-------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------|--------|--------|-------|
| Biratnagar                                                                                |          |                                                                                    |        |        |       |
| Home                                                                                      | About PU | Faculties                                                                          | Result | Notice | Links |
| <b><u>Notice</u></b><br>Admission open<br>Result of MCA 1st published<br>New Affiliations |          | <b><u>Message From VC</u></b><br>Welcome to Purbanchal University<br>{Placeholder} |        |        |       |
| All rights reserved © Purbanchal University                                               |          |                                                                                    |        |        |       |

Clicking on Result should open a page on {Placeholder} with following fields:

- (a) Full name
- (b) Roll No.
- (c) Date of Birth
- (d) Registration No.

Also perform proper validation of control on the Result page.

Contd. ...

**Answer SIX questions.**

4. How is it possible for applications written in various languages to compile and run in a common dot net framework? Explain.
5. Explain event driven programming in detail.
6. Explain the architecture of ADO.Net. Differentiate between ADO and ADO.Net. 3+3
7. Discuss and explain the concept of DataGridView deploying application.
8. Elaborate the problems of window pollution and ways to remove them.
9. Explain master and target objects. Discuss about the mechanism of drag and drop taking into account the various phases of direct manipulations. 2+4
10. Write short notes on Any TWO: 2×3=6
  - (a) Polymorphism
  - (b) Session
  - (c) CUI v/s GUI

